



JISHIN BIJUMON GEORGE

ELECTRONICS AND COMMUNICATION ENGINEER | EMBEDDED SYSTEMS & PCB DESIGN ENGINEER

- +91 9037360398
- jishinbiju004@gmail.com
- Kottarakara, Kerala, India, 691557
- [linkedin.com/in/jishinbijumon](https://www.linkedin.com/in/jishinbijumon)
- github.com/JishinBijumon

EDUCATION

- B.Tech in Electronics and Communication Engineering**
2022-26 | Amal Jyothi College of Engineering, Kottayam, Kanjirapally | KTU
- Higher Secondary Education**
2021-22 | International Indian School, Dammam, Saudi Arabia | CBSE |
- Secondary Education**
2019-20 | International Indian School, Dammam, Saudi Arabia | CBSE |

TECHNICAL SKILLS

- PCB Design & EDA Tools:**
Altium Designer, KiCad, OrCAD
- Embedded Platforms:**
ESP32, ESP8266, Arduino
- Embedded & Hardware Skills:**
Sensor Interfacing, PCB Debugging, Hardware Testing
- Programming & Development Tools:**
C and Python (basics), Arduino IDE, GitHub
- IoT & Edge Technologies:**
Firebase, Supabase, Edge Impulse
- Professional Skills:**
Problem Solving, Analytical Thinking, Time Management

CERTIFICATION & PUBLICATION

- Research Publication: Smart Specs for Visually Impaired People using ESP32 – MAT Journals Pvt. Ltd | Vol. 2, Issue 2 | 2025
- Industry Immersion Program on Embedded Systems NeST Digital | June 2025
- Embedded Systems Internship (Arduino IDE) – Amal Jyothi College of Engineering (Autonomous)

ABOUT ME

Electronics and Communication Engineering graduate with hands-on experience in PCB design, embedded systems, and IoT development. Proficient in Altium Designer, KiCad, ESP32, and Arduino-based system development. Experienced in schematic capture, PCB layout, hardware prototyping, and embedded application development.

INTERNSHIP EXPERIENCE

- NeST DIGITAL, ERNAKULAM**
Academic Internship
IoT and Embedded Systems - June 2025 (1 Month)
- AMAL JYOTHI COLLEGE OF ENGINEERING, (AUTONOMOUS) KOTTAYAM, KANJIRAPALLY, 686518**
Academic Internship
Embedded Systems Internship (Arduino IDE) (Department-organized) - May- June 2024 (1 Month)

PROJECTS

EMBEDDED / IoT PROJECTS:

- BRAIN-CONTROLLED SMART WHEEL CHAIR FOR PARALLIZED PATIENTS USING EEG SIGNALS AND LINE-FOLLOWING**
Developing brain-controlled smart wheelchair using EEG signal acquisition for intent detection and line-following navigation for assisted mobility.
- SMART SPECS FOR VISUALLY IMPAIRED PEOPLE**
Designed smart eyewear for visually impaired users using ESP32-CAM and sensors with audio feedback for basic obstacle awareness.
- HUMAN ELDERLY MONITORING SYSTEM**
Developed a compact chest-wearable IoT-based elderly monitoring system with real-time health tracking, fall detection, GPS/GSM alerts, and multi-sensor integration using ESP32.
- IOT-BASED GAS LEAKAGE DETECTION AND HOME SAFETY MONITORING SYSTEM**
Developed an IoT-based home safety system to detect gas leaks and alert users in real time.

HARDWARE PROJECTS :

- CUSTOM PCB BOARD FOR BCI- BASED WHEELCHAIR (FINAL YEAR PROJECT)**
Designed a custom PCB for EEG-based wheelchair control using KiCad, integrating power regulation, signal conditioning, and motor control interfaces.
- 5V TO 3.3V VOLTAGE REGULATOR**
Developed a PCB for stable power regulation in embedded applications.
- UART CONVERTER USING CP340G**
Designed a module for reliable serial communication between devices.
- MOTOR DRIVER MODULE**
Designed a motor driver module PCB for DC motor control in embedded systems.
- ARDUINO UNO DEVELOPMENT BOARD**
Currently designing an Arduino Uno-compatible development board, focusing on full-board schematic capture and PCB layout.

REFERENCE:

- AGI JOSEPH GEORGE**
Assistant Professor- ECE
Amal Jyothi College of Engineering (AUTONOMOUS), Kottayam, Kanjirapally, 686518
agijosephgeorge@amaljyothi.ac.in
- DR. GEEVARGHESE TITUS**
Professor, HoD- ECE
Amal Jyothi College of Engineering (AUTONOMOUS), Kottayam, Kanjirapally, 686518
geevarghesetitus@amaljyothi.ac.in